SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Code of Conduct:

*I **Yuvansri A R (192421029)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Scenario-Based Questions

1. Low Brightness Image

An image captured in low light appears very dark with poor visibility.
(a) Interpret the scenario and explain the cause of low brightness.
(b) Identify key issues affecting image visibility.
(c) Apply an appropriate enhancement technique.
(d) Justify your choice with reasoning.
(e) Present your explanation clearly.

ANSWER

(a) Interpret the Scenario and Explain the Cause of Low Brightness

A low-brightness image is one that appears **very dark**, making objects difficult to identify. This usually occurs when the image is captured under **insufficient lighting conditions**, such as at night, indoors with poor illumination, or when the camera uses a short exposure time or low sensor sensitivity (ISO).

Causes of low brightness:

- Poor or insufficient lighting
- Low camera exposure settings
- Low sensor sensitivity (ISO)
- Shadows covering important regions
- Camera lens allowing limited light

(b) Key Issues Affecting Image Visibility

The major problems caused by low brightness include:

- **Poor visibility** of objects and scene details
- **Loss of fine details** in dark regions
- **Low contrast** between objects and background
- **Difficulty in edge and feature detection**
- **Reduced accuracy** in computer vision tasks such as object detection and recognition

(c) Appropriate Enhancement Technique

Histogram Equalization (or Adaptive Histogram Equalization - CLAHE)

Steps:

1. Read the low-brightness image.
2. Convert it to grayscale (if required).
3. Apply **Histogram Equalization** to spread the intensity values across the full brightness range.

4. For images with uneven lighting, use **CLAHE (Contrast Limited Adaptive Histogram Equalization)** for better local enhancement.
5. Display the enhanced image.

(d) Justification

Histogram Equalization is suitable because it:

- Increases the overall brightness of the image.
- Enhances contrast by redistributing pixel intensity values.
- Reveals hidden details in dark regions.
- Improves object visibility and feature extraction.
- Produces a clearer image for further image-processing operations.

For images with non-uniform illumination, **CLAHE** is preferred because it enhances local contrast while preventing excessive noise amplification.

(e) Clear and Structured Explanation

Aspect	Explanation
Scenario	Image appears very dark due to insufficient lighting.
Cause	Low illumination, improper exposure, or low camera sensitivity.
Issues	Poor visibility, loss of details, low contrast, difficult edge detection.
Technique Used	Histogram Equalization (or CLAHE for better local enhancement).
Reason	Improves brightness and contrast, making hidden details visible and enhancing overall image quality.

2. Overexposed Image

An image appears excessively bright, losing important details.
(a) Interpret the scenario and explain overexposure.
(b) Identify key issues affecting intensity representation.
(c) Apply a suitable correction technique.
(d) Justify your choice logically.
(e) Present your explanation in a structured manner.

ANSWER

(a) Interpret the Scenario and Explain Overexposure

An **overexposed image** is an image that appears **too bright**, causing bright areas to become white and lose important details. Overexposure occurs when **too much light reaches the camera sensor**, exceeding its ability to accurately record intensity values.

Causes of overexposure:

- Excessive lighting conditions
- Long camera exposure time
- High ISO sensitivity
- Large aperture (low f-number)
- Incorrect camera exposure settings

(b) Key Issues Affecting Intensity Representation

The major problems caused by overexposure are:

- **Loss of details** in bright regions (highlight clipping)
- **Pixel intensity saturation** (many pixels become 255 in an 8-bit image)
- **Reduced contrast** between bright objects
- **Washed-out appearance**
- **Difficulty in feature extraction, edge detection, and object recognition**

(c) Suitable Correction Technique

Gamma Correction ($\gamma > 1$)

Steps:

1. Read the overexposed image.
2. Normalize pixel values between 0 and 1.
3. Apply **Gamma Correction** using a gamma value greater than 1 (e.g., $\gamma = 2.0$).
4. Convert the image back to the original intensity range (0–255).
5. Display the corrected image.

Alternative techniques:

- Intensity (Brightness) Adjustment
- Contrast Stretching
- Histogram Specification (when a reference image is available)

(d) Justification

Gamma Correction is appropriate because it:

- Reduces excessive brightness without significantly affecting darker regions.
- Restores details in moderately bright areas.
- Improves overall contrast and visual quality.
- Produces a more natural-looking image.
- Is computationally simple and widely used in image enhancement.

If the highlights are **completely saturated (pure white)**, the lost information cannot be fully recovered because those intensity values are permanently clipped.

(e) Structured Explanation

Aspect	Explanation
Scenario	Image appears excessively bright, making important details difficult to see.
Cause	Too much light reaches the camera sensor due to incorrect exposure settings or strong illumination.
Issues	Highlight clipping, intensity saturation, washed-out appearance, reduced contrast, loss of details.
Technique Used	Gamma Correction ($\gamma > 1$) to reduce brightness and improve contrast.
Reason	Compresses high-intensity values, restores visual balance, and enhances image quality.

3. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.
(a) Interpret the scenario and describe Gaussian noise.
(b) Identify key issues affecting quality.
(c) Apply an appropriate smoothing filter.
(d) Justify your method over alternatives.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

Gaussian noise is a **random noise** that follows a **normal (Gaussian) distribution**, causing small variations in pixel intensity. It mainly affects smooth regions, making the image appear grainy and reducing visual quality.

(b) Key Issues:

The major issues are **reduced image clarity**, blurred fine details, lower signal-to-noise ratio (SNR), and decreased accuracy in image analysis and feature extraction.

(c) Appropriate Smoothing Filter:

The most suitable filter is the **Gaussian Filter**, which smooths the image by reducing Gaussian noise while preserving the overall image structure.

(d) Justification:

The **Gaussian Filter** is specifically designed for Gaussian noise. It provides smooth noise reduction with less distortion than a Mean Filter and is more suitable than a Median Filter, which is mainly used for salt-and-pepper noise.

(e) Structured Explanation:

When an image is affected by **Gaussian noise**, the **Gaussian Filter** is the preferred choice because it effectively reduces random intensity variations while maintaining the natural appearance of the image. This improves image quality and supports accurate image processing tasks.

4. Speckle Noise

An image obtained from ultrasound imaging shows speckle noise.
(a) Interpret the scenario and explain speckle noise characteristics.
(b) Identify key issues affecting clarity.
(c) Apply a suitable filtering technique.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

Speckle noise is a **multiplicative noise** commonly found in **ultrasound, SAR, and radar images**. It appears as a grainy pattern that reduces image quality and makes tissue structures difficult to distinguish.

(b) Key Issues:

The major issues are **reduced image clarity**, loss of fine details, poor contrast, and difficulty in identifying tissue boundaries and abnormalities.

(c) Suitable Filtering Technique:

The most suitable technique is the **Lee Filter** (or **Frost Filter**), which is specifically designed to reduce speckle noise while preserving important edges and details.

(d) Justification:

The **Lee Filter** is preferred because it effectively reduces multiplicative speckle noise while preserving edges and fine structures. It performs better than Mean or Gaussian filters, which tend to blur important image details.

(e) Structured Explanation:

For ultrasound images affected by **speckle noise**, the **Lee Filter** is the most appropriate method. It reduces grainy noise, preserves edges and tissue details, and improves image clarity, making it suitable for accurate medical image analysis.

5. Loss of Fine Details

An image loses fine details after processing.
(a) Interpret the scenario and explain detail loss.
(b) Identify key issues related to filtering.
(c) Apply a suitable enhancement technique.
(d) Justify your method with reasoning.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

After image processing, fine details such as edges, textures, and small features are lost due to excessive smoothing or filtering. This reduces image sharpness and important visual information.

(b) Key Issues:

The major issues are blurred edges, loss of texture, reduced sharpness, and lower accuracy in feature extraction and object recognition.

(c) Suitable Enhancement Technique:

The most suitable technique is Unsharp Masking (Image Sharpening) to restore fine details and enhance edge clarity.

(d) Justification:

Unsharp Masking enhances edges and fine details without significantly affecting the overall image. It is more effective than applying additional smoothing filters, which would further reduce image details.

(e) Structured Explanation:

When an image loses fine details after processing, Unsharp Masking is the preferred enhancement technique. It restores edge sharpness, improves texture visibility, and enhances overall image quality, making the image suitable for accurate analysis and interpretation.

6. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.
(a) Interpret the scenario and explain the need for edge enhancement.
(b) Identify key issues in detecting edges.
(c) Apply an appropriate method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The application requires **highlighting object edges** so that important features can be extracted accurately. Edge enhancement improves the visibility of boundaries, making feature extraction easier.

(b) Key Issues:

The major issues are **image noise**, weak or blurred edges, low contrast, and false edge detection, which reduce the accuracy of feature extraction.

(c) Appropriate Method:

The most suitable method is **Sobel Edge Detection** (or **Canny Edge Detection** for higher accuracy).

(d) Justification:

Canny Edge Detection is preferred because it reduces noise, detects both strong and weak edges, and produces thin, continuous edges. This improves the accuracy of feature extraction compared to simpler methods.

(e) Structured Explanation:

For applications requiring edge enhancement, **Canny Edge Detection** is the most effective method. It enhances object boundaries, minimizes the effect of noise, and provides clear, continuous edges, making it ideal for accurate feature extraction.

7. Background Noise Removal

An image contains background noise affecting segmentation accuracy.
(a) Interpret the scenario and explain noise impact.
(b) Identify key issues affecting segmentation.
(c) Apply a suitable preprocessing method.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpret the scenario and explain noise impact

Background noise refers to unwanted random variations in an image that do not belong to the actual object or scene. It may be caused by poor lighting, sensor errors, transmission interference, or environmental conditions.

Impact on segmentation:

- Produces false object boundaries.
- Creates unwanted regions during segmentation.
- Reduces image quality and object visibility.
- Increases segmentation errors and decreases accuracy.

(b) Identify key issues affecting segmentation

The major issues caused by background noise are:

- **False edges:** Noise creates artificial edges that confuse segmentation algorithms.
- **Loss of object details:** Important features may be hidden or distorted.
- **Over-segmentation:** One object may be divided into multiple regions.
- **Under-segmentation:** Different objects may merge into one region.
- **Reduced accuracy:** Noise leads to incorrect object detection and classification.

(c) Apply a suitable preprocessing method

Suitable Method: Median Filtering

Median filtering is an effective preprocessing technique for removing background noise, especially salt-and-pepper noise, while preserving edges.

Steps:

1. Input the noisy image.
2. Apply a median filter (e.g., 3×3 or 5×5 kernel).
3. Replace each pixel with the median value of its neighboring pixels.
4. Use the filtered image for segmentation.

Example (OpenCV):

```
import cv2
# Read imageimg = cv2.imread("noisy_image.jpg")
# Apply Median Filterfiltered = cv2.medianBlur(img, 5)
# Save resultcv2.imwrite("filtered_image.jpg", filtered)
```

(d) Justify your approach logically

Median filtering is preferred because:

- It effectively removes impulse (salt-and-pepper) noise.
- It preserves object edges better than averaging filters.
- It improves the quality of segmented regions.
- It reduces false boundaries and isolated noisy pixels.
- It is simple, computationally efficient, and widely used as a preprocessing step before segmentation.

Hence, median filtering increases segmentation accuracy while maintaining important image features.

(e) Present your explanation clearly

Aspect	Explanation
Problem	Background noise reduces segmentation accuracy.
Impact	Produces false edges, missing details, and incorrect object boundaries.
Key Issues	Over-segmentation, under-segmentation, reduced accuracy, false regions.
Preprocessing Method	Median Filtering
Reason for Selection	Removes noise while preserving edges and important image details.
Outcome	Cleaner image with improved segmentation accuracy and reliable object detection.

8. Threshold Selection Problem

An image requires segmentation, but selecting a threshold is difficult.
(a) Interpret the scenario and explain the challenge.
(b) Identify key issues in threshold selection.
(c) Apply a suitable thresholding method.
(d) Justify your approach logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image requires **segmentation**, but choosing the correct threshold is difficult because the object and background do not have a clear intensity difference. This makes accurate separation challenging.

(b) Key Issues:

The major issues are **overlapping intensity values**, uneven illumination, image noise, and poor contrast, which can lead to incorrect segmentation.

(c) Suitable Thresholding Method:

The most appropriate method is **Otsu's Thresholding**, which automatically determines the optimal threshold value.

(d) Justification:

Otsu's Thresholding is preferred because it automatically selects the threshold that best separates the foreground and background by maximizing the variance between the two classes. It is more accurate and reliable than manually choosing a fixed threshold.

(e) Structured Explanation:

When threshold selection is difficult, **Otsu's Thresholding** provides an automatic and efficient solution. It improves object-background separation, reduces manual effort, and produces accurate segmentation for many grayscale images.

9. Texture-Based Segmentation

An image contains regions distinguishable by texture rather than intensity.
(a) Interpret the scenario and explain segmentation challenges.
(b) Identify key issues with intensity-based methods.
(c) Apply an appropriate segmentation approach.
(d) Justify your method logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image contains regions that differ mainly in **texture** rather than intensity. Since the intensity values are similar, segmentation based only on brightness is difficult.

(b) Key Issues:

The major issues are **similar intensity values**, overlapping gray levels, complex texture patterns, and poor performance of intensity-based segmentation methods.

(c) Appropriate Segmentation Approach:

The most suitable approach is **Texture-Based Segmentation using GLCM (Gray Level Co-occurrence Matrix)** or **Local Binary Pattern (LBP)** features, followed by clustering (e.g., K-Means).

(d) Justification:

GLCM/LBP extracts texture features such as contrast, homogeneity, and correlation instead of relying only on intensity. This allows accurate separation of regions with similar brightness but different textures.

(e) Structured Explanation:

When regions differ by **texture rather than intensity**, **Texture-Based Segmentation (GLCM/LBP)** is the best approach. It analyzes texture patterns, accurately separates similar-intensity regions, and provides better segmentation than intensity-based methods.

10. Multiple Object Detection

An image contains multiple objects with similar intensity values.
(a) Interpret the scenario and explain segmentation difficulty.
(b) Identify key issues in distinguishing objects.
(c) Apply a suitable segmentation technique.
(d) Justify your choice logically.
(e) Present your explanation clearly.

ANSWER

(a) Interpretation of the Scenario:

The image contains **multiple objects with similar intensity values**, making it difficult to distinguish one object from another using only pixel brightness. As a result, simple segmentation methods may merge multiple objects into a single region or fail to identify individual objects correctly.

(b) Key Issues:

The major challenges include **similar intensity values**, overlapping or touching objects, weak object boundaries, image noise, and low contrast. These issues reduce the accuracy of object detection and segmentation.

(c) Suitable Segmentation Technique:

The most suitable technique is **Watershed Segmentation**. This method treats the image as a topographic surface and uses intensity gradients to identify and separate connected or overlapping objects. Morphological operations can be applied before the watershed algorithm to improve the results.

(d) Justification:

Watershed Segmentation is preferred because it uses **gradient information** rather than only intensity values. It accurately separates touching or overlapping objects, preserves object boundaries, and performs better than simple thresholding when objects have similar brightness. This makes it suitable for medical, industrial, and biological image analysis.

(e) Structured Explanation:

When an image contains **multiple objects with similar intensity values**, simple thresholding cannot separate them effectively. **Watershed Segmentation** analyzes gradient information to detect object boundaries and divide connected regions into individual objects. This approach improves segmentation accuracy, preserves object shapes, and enables reliable object detection and analysis

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand